MA1521 L1 Week 08 Mon 20261005

Evaluate each of the following iterated integrals:

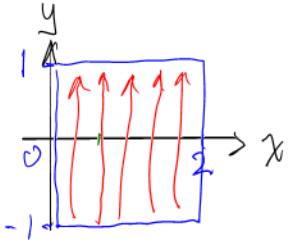
a) $\displaystyle\int_0^2 \int_{-1}^1 (6x^2 + 2y)\, dy\, dx$

$$\int_{x=0}^{x=2} \left(\int_{y=-1}^{y=1} (6x^2 + 2y)\, dy \right) dx$$

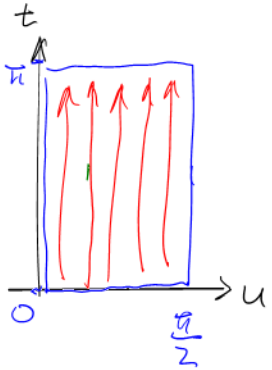
$$= \int_{x=0}^{x=2} \Big(\big[6x^2 y + y^2 \big]_{y=-1}^{y=1} \Big)\, dx$$

$$= \int_{x=0}^{x=2} \Big(\big(6x^2 \cdot 1 + 1^2 \big) - \big(6x^2 (-1) + (-1)^2 \big) \Big)\, dx$$

$$= \int_{x=0}^{x=2} 12x^2\, dx = \big[4x^3 \big]_{x=0}^{x=2} = 32$$



b) $\displaystyle\int_0^{\pi/2} \int_0^{\pi} (u \sin t + t \cos u)\, dt\, du$



$$\int_{u=0}^{u=\pi/2} \left(\int_{t=0}^{t=\pi} (u \sin t + t \cos u)\, dt \right) du$$

$$= \int_{u=0}^{u=\pi/2} \Big(\big[-u \cos t + \tfrac{t^2}{2} \cos u \big]_{t=0}^{t=\pi} \Big)\, du$$

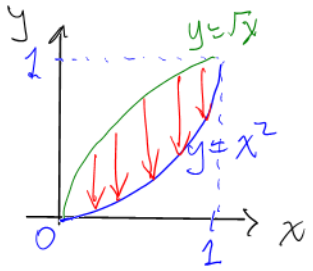
$$= \int_{u=0}^{u=\pi/2} \Big(\big(u + \tfrac{\pi^2}{2} \cos u \big) - \big(-u + 0 \big) \Big)\, du$$

$$= \int_{u=0}^{u=\pi/2} \Big(2u + \tfrac{\pi^2}{2} \cos u \Big)\, du$$

$$= \Big[u^2 + \tfrac{\pi^2}{2} \sin u \Big]_{u=0}^{u=\pi/2} = \tfrac{\pi^2}{4} + \tfrac{\pi^2}{2} = \tfrac{3\pi^2}{4}$$

Evaluate each of the following iterated integrals:

c) $\displaystyle\int_0^1 \left(\int_{\sqrt{x}}^{x^2} 2x^2 y \, dy \right) dx$



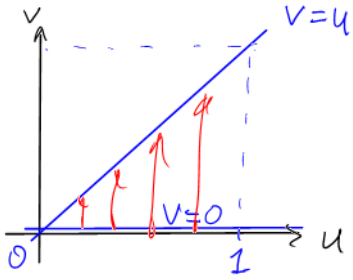
$$\int_{x=0}^{x=1} \left(\int_{y=\sqrt{x}}^{y=x^2} 2x^2 y \, dy \right) dx$$

$$= \int_{x=0}^{x=1} \left[2x^2 \frac{y^2}{2} \right]_{y=\sqrt{x}}^{y=x^2} dx$$

$$= \int_{x=0}^{x=1} \left[x^2 x^4 - x^2 \cdot x \right] dx \qquad (x^6 - x^3)$$

$$= \left[\frac{x^7}{7} - \frac{x^4}{4} \right]_{x=0}^{x=1} = \frac{1}{7} - \frac{1}{4} = \frac{-3}{28}$$ //

d) $\displaystyle\int_0^1 \int_0^u \sqrt{u^2+4} \, dv \, du$



$$\int_{u=0}^{u=1} \left(\int_{v=0}^{v=u} \sqrt{u^2+4} \, dv \right) du$$

$$= \int_{u=0}^{u=1} \left[\sqrt{u^2+4} \cdot v \right]_{v=0}^{v=u} du$$

$$= \int_{u=0}^{u=1} \sqrt{u^2+4} \cdot u \, du$$

$$= \frac{1}{3}\left(u^2+4\right)^{3/2} \Big|_{u=0}^{u=1} = \frac{5\sqrt{5}-8}{3}$$ //

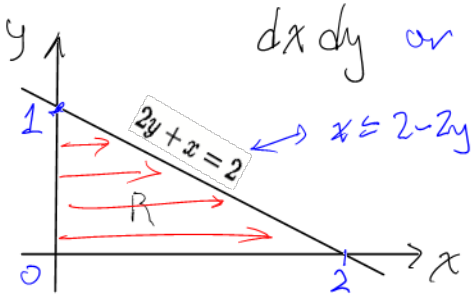
Evaluate each of the following double integrals over the indicated region R.

a) $\iint_R x\,dA$; R is the finite region bounded by the axes and $2y + x = 2$

$x = 0$

$y = 0$

$dA = dx\,dy$ or $dy\,dx$



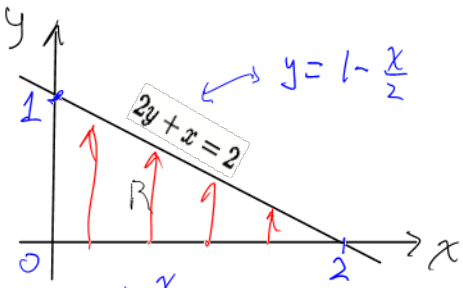
(1) $\iint_R x\,dA = \int_{y=0}^{y=1} \left(\int_{x=0}^{x=2-2y} x\,dx \right) dy$

$= \int_{y=0}^{y=1} \left(\frac{x^2}{2} \Big|_{x=0}^{x=2-2y} \right) dy$

$= \int_{y=0}^{y=1} \frac{(2-2y)^2}{2}\,dy = 2\int_{y=0}^{y=1} 1 - 2y + y^2\,dy$

$= 2\left[y - y^2 + \frac{y^3}{3} \right]_{y=0}^{y=1}$

$= 2\left[1 - 1 + \frac{1}{3} \right] = \frac{2}{3}$ //



(2) $\iint_R x\,dA = \int_{x=0}^{x=2} \left(\int_{y=0}^{y=1-\frac{x}{2}} x\,dy \right) dx$

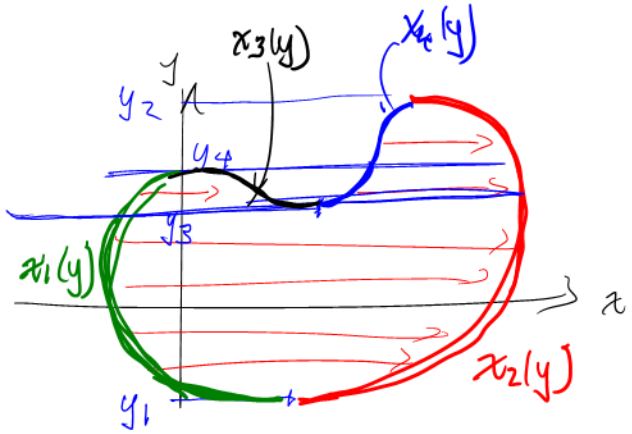
$= \int_{x=0}^{x=2} x \cdot \left(1 - \frac{x}{2}\right) dx$

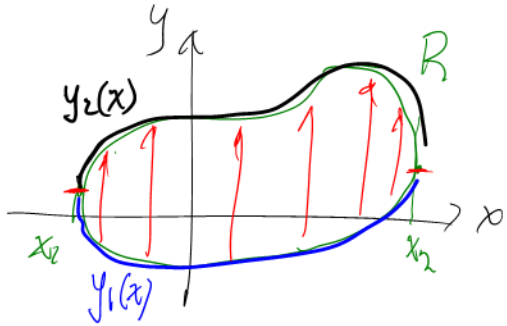
$= \left[\frac{x^2}{2} - \frac{x^3}{6} \right]_{x=0}^{x=2}$

$= \frac{4}{2} - \frac{8}{6} = \frac{2}{3}$ //

Fubini-Tonelli Thm: If $f(x,y)$ is continuous, and R is closed bounded region in xy-plane, then

$$\iint_R f(x,y)\,dA = \int_{y=y_1}^{y=y_2}\left(\int_{x=x_1(y)}^{x=x_2(y)} f(x,y)\,dx\right)dy = \int_{x=x_1}^{x=x_2}\left(\int_{y=y_1(x)}^{y=y_2(x)} f(x,y)\,dy\right)dx$$

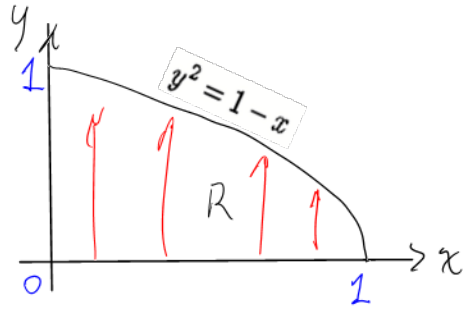




$$\int_{y=y_1}^{y=y_3}\left(\int_{x=x_1(y)}^{x=x_2(y)} -\,dx\right)dy + \int_{y=y_3}^{y=y_4}\left(\int_{x=x_1(y)}^{x=x_3(y)} -\,dx + \int_{x=x_4(y)}^{x=x_2(y)} -\,dx\right)dy + \int_{y=y_4}^{y=y_2}\left(\int_{x=x_4(y)}^{x=x_2(y)} -\,dx\right)dy$$

Evaluate each of the following double integrals over the indicated region R.

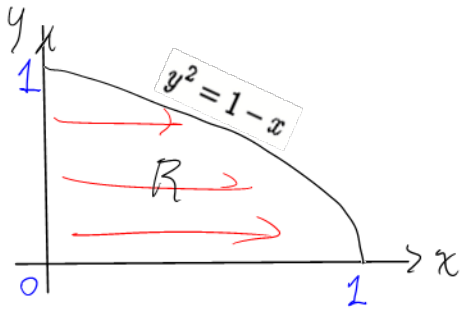
b) $\iint_R (2x + y^2)\, dA$; R is the finite region in the first quadrant bounded by the axes and $y^2 = 1 - x$



$$\iint_R (2x + y^2)\, dA$$

$$= \int_{x=0}^{x=1} \left(\int_{y=0}^{y=\sqrt{1-x}} (2x + y^2)\, dy \right) dx$$

Exercise: do this, check that you got $\frac{2}{3}$.



$$\iint_R (2x + y^2)\, dA$$

$$= \int_{y=0}^{y=1} \left(\int_{x=0}^{x=1-y^2} (2x + y^2)\, dx \right) dy$$

$$= \int_{y=0}^{y=1} \left[x^2 + xy \right]_{x=0}^{x=1-y^2} dy$$

$$= \int_{y=0}^{y=1} \underbrace{(1-y^2)^2 + (1-y^2)y}_{1-y^2}\, dy$$

$$= \left(y - \frac{y^3}{3} \right)_{y=0}^{y=1}$$

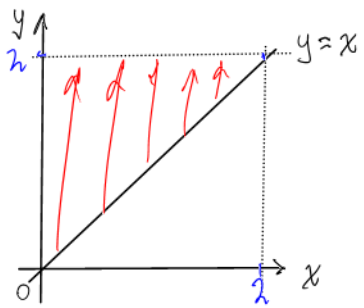
$$= 1 - \frac{1}{3} = \frac{2}{3}$$

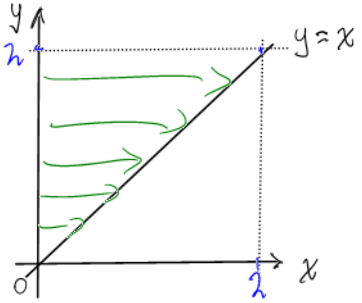
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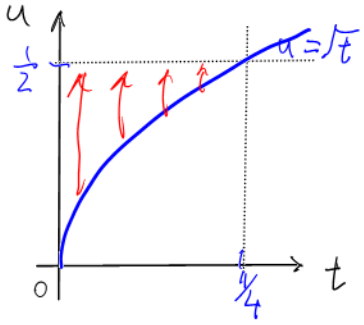
Fubini - Tonelli Thm

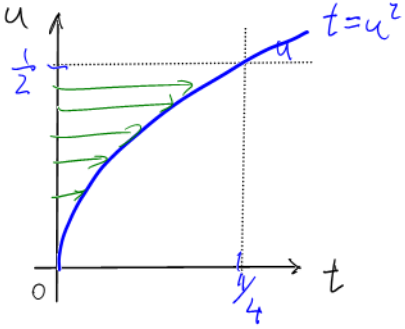
Evaluate each of the following iterated integrals, by changing the order of integration

a) $\displaystyle\int_0^2 \int_x^2 e^{-y^2}\,dy\,dx$ b) $\displaystyle\int_0^{1/4} \int_{\sqrt{t}}^{1/2} \frac{e^u}{u}\,du\,dt$ c) $\displaystyle\int_0^1 \int_{x^{1/3}}^1 \frac{1}{1+u^4}\,du\,dx$









a) $\displaystyle\int_0^2 \int_x^2 e^{-y^2}\,dy\,dx$ swap order $= \int_{y=0}^{y=2} \int_{x=0}^{x=y} e^{-y^2}\,dx\,dy$

$= \displaystyle\int_{y=0}^{y=2} e^{-y^2} \cdot y\,dy$

$= \left[\frac{-1}{2} e^{-y^2} \right]_{y=0}^{y=2}$

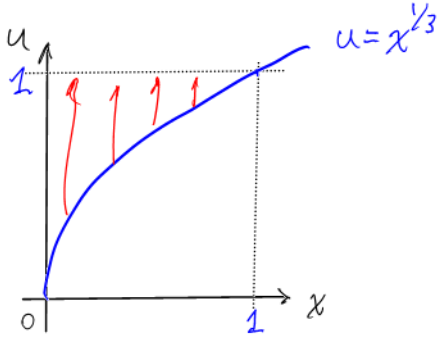
$= \frac{1}{2}(1 - e^{-4})$

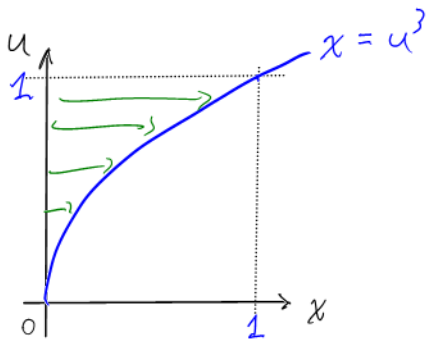
b) $\displaystyle\int_0^{1/4} \int_{\sqrt{t}}^{1/2} \frac{e^u}{u}\,du\,dt = \int_{u=0}^{u=\frac{1}{2}} \int_{t=0}^{t=u^2} \frac{e^u}{u}\,dt\,du$

$= \displaystyle\int_{u=0}^{u=\frac{1}{2}} \underbrace{\frac{e^u}{u} u^2}_{u e^u}\,du$

($\int$ by parts) $= \left[(u-1)e^u \right]_{u=0}^{u=\frac{1}{2}}$

$= 1 - \frac{\sqrt{e}}{2}$ #





c) $$\int_0^1 \int_{x^{1/3}}^1 \frac{1}{1+u^4}\, du\, dx$$

Ch. order

$$= \int_{u=0}^{u=1} \int_{x=0}^{x=u^3} \frac{1}{1+u^4}\, dx\, du$$

$$= \int_{u=0}^{u=1} \frac{1}{1+u^4}\, u^3\, du$$

$$= \left[\frac{1}{4} \ln(1+u^4) \right]_{u=0}^{u=1} = \frac{\ln 2}{4}$$